

Problem 2 (Uphill paths), Cell 2: $U(Q_5)$

Team submission

BAINSA Hackathon, 26 September 2026

1 Cell 2

Official setting. As in Cell 1 (§A.1): $U(G)$ is the least number of uphill paths over all labellings of G . Values are handed in together with an explicit labelling, listed as the 2^d vertices in increasing label order.

Task. Determine $U(Q_5)$, and give a labelling attaining it.

Status. Solved. The upper bound is by hand. The lower bound is **computer-assisted**: one SAT computation, below one second, repeated with two solvers.

Answer. $U(Q_5) = 88$, and $\nabla(Q_5) = 14$ (Theorem 1.2). The labelling is in §1.2.

1.1 The decycling number of Q_5

The counting bound of Lemma A.7 gives only $4\nabla(Q_5) \geq 3 \cdot 16 + 1$, that is, $\nabla(Q_5) \geq 13$, and so $U(Q_5) \geq 84$. This bound is not attained.

Lemma 1.1 (computer-assisted). Q_5 has no decycling set with 13 or fewer vertices. Hence $\nabla(Q_5) \geq 14$.

Method, and why the computation proves the lemma. This is claim (L5) of `code/uphill_lower.py`. The variables are x_v ($v \in V(Q_5)$), with x_v meaning $v \in R$. The formula F contains:

- (a) a CNF encoding (sequential counter) of $\sum_v x_v \leq 13$;
- (b) for every 4-cycle K of Q_5 , the clause $\bigvee_{v \in K} x_v$;
- (c) the clauses $\bigvee_{v \in K} x_v$ for cycles K found during the loop below.

Each clause of type (b) or (c) says that R meets the cycle K , and every decycling set does this. So every decycling set of size at most 13 satisfies F at every stage.

Loop. The solver returns a model R , or reports UNSAT. If it returns R , we compute $Q_5 - R$ with `networkx`. If this graph has a cycle, we add the clauses of a cycle basis of every cyclic component (type (c)) and ask again. The new clauses are violated by the current model, so no model is returned twice. Hence the loop ends, after finitely many rounds. It ends in one of two ways: a model R for which $Q_5 - R$ is a forest (a decycling set of size ≤ 13), or UNSAT.

Result. UNSAT, after 348 rounds in 0.4s with CaDiCaL 1.5.3, and after 357 rounds in 0.3s with Glucose 4, both through `python-sat`. Since every decycling set of size ≤ 13 would be a model, none exists. The only facts used are the correctness of the cardinality encoding and of the SAT solvers. Running two independent solvers guards against a solver bug. $\square$

Theorem 1.2. $\nabla(Q_5) = 14$ and $U(Q_5) = 88$.

Proof. Lower bound: Lemma 1.1 and Theorem A.4 give $U(Q_5) \geq 32 + 4 \cdot 14 = 88$. Upper bound: $C = \{00000, 11110\}$ lies in E_5 and has distance 4. By Proposition A.6, $E_5 \setminus C$ is a decycling set of size $16 - 2 = 14$, and the labelling L_C has $32 + 4 \cdot 14 = 88$ uphill paths. $\square$

Remark 1.3. A set $C \subseteq E_5$ with pairwise distances ≥ 4 has at most two elements. Suppose it had three. Translating by one of them (an even vector, which preserves E_5 and all distances), we may assume $C = \{0, a, b\}$. Then $|a| = |b| = 4$, so a and b are 4-subsets of a 5-set and meet in at least 3 coordinates. This gives $|a \oplus b| \leq 2$, a contradiction. So Proposition A.6 cannot give less than 88, and Lemma 1.1 shows that no labelling can.

1.2 The labelling

L_C with $C = \{00000, 11110\}$: codewords, then the 16 odd vertices, then the remaining 14 even vertices, each block in increasing binary order. It has 88 uphill paths (checked with `code/uphill_eval.py`).

```
1: 00000 11110 00001 00010 00100 00111 01000 01011
9: 01101 01110 10000 10011 10101 10110 11001 11010
17: 11100 11111 00011 00101 00110 01001 01010 01100
25: 01111 10001 10010 10100 10111 11000 11011 11101
```

A Results used: Cell 1 ($U(Q_3)$ and $U(Q_4)$), as handed in for that cell

Official setting.

Let G be a finite simple graph on n vertices. A *labelling* of G is a bijection f from $V(G)$ to $\{1, 2, \dots, n\}$. Fix a labelling. A vertex v is a *valley* if every neighbour w of v has $f(w) > f(v)$; an isolated vertex counts as a valley. An *uphill path* is a sequence $(v_1, v_2, \dots, v_k)$ with $k \geq 1$ in which v_1 is a valley, v_i and v_{i+1} are adjacent for each i , and $f(v_1) < f(v_2) < \dots < f(v_k)$. A valley on its own ($k = 1$) counts as an uphill path. Let $U(G)$ be the smallest possible number of uphill paths, taken over all labellings of G . (IMO 2022 Problem 6 asks for U of the $n \times n$ grid graph; the answer is $2n^2 - 2n + 1$.) The d -dimensional hypercube Q_d has vertex set $\{0, 1\}^d$, and two vertices are adjacent when they differ in exactly one coordinate; Q_d has 2^d vertices and $d \cdot 2^{d-1}$ edges. For C1 to C4, hand in the values and, for each value, an explicit labelling that attains it, as a list of the 2^d vertices in increasing label order, written as 0/1 strings.

Task. Determine $U(Q_3)$ and $U(Q_4)$, and give a labelling attaining each value.

Status. **Solved**, by hand. No computation is needed; the labellings were also counted mechanically (`code/uphill_eval.py`).

Answer and where it is proved.

- $U(Q_3) = 14$ and $U(Q_4) = 34$ (Theorem A.8); labellings in §A.8.
- The proof rests on an exact formula for the number of uphill paths (Proposition A.3). It gives $U(Q_d) \geq 2^d + (d-1)\nabla(Q_d)$ for every d , where ∇ is the decycling number (Theorem A.4). In the other direction, every even-weight code with minimum distance 4 gives a labelling (Proposition A.6).
- Sections A.1–A.6 hold for every d and are used by all the later cells of this problem.

A.1 Counting uphill paths

Fix a graph G and a labelling f . A *lower neighbour* of v is a neighbour w with $f(w) < f(v)$, and a *higher neighbour* is a neighbour w with $f(w) > f(v)$. Write $\text{out}(v)$ for the number of higher neighbours of v . So v is a valley if and only if it has no lower neighbour. Let $N(v) = N_f(v)$ be the number of uphill paths whose last vertex is v , and $P(f) = \sum_v N(v)$ the number of uphill paths. Then $U(G) = \min_f P(f)$.

Lemma A.1 (Recursion). *$N(v) = 1$ if v is a valley, and $N(v) = \sum_w N(w)$, summed over the lower neighbours w of v , otherwise. In particular $N(v) \geq 1$ for every v .*

Proof. If $(v_1, \dots, v_k)$ is an uphill path with $v_k = v$ and $k \geq 2$, then v_{k-1} is a lower neighbour of v and $(v_1, \dots, v_{k-1})$ is an uphill path ending at v_{k-1} . Conversely, if w is a lower neighbour of v , then every uphill path ending at w extends by v to an uphill path ending at v . These two maps are inverse to each other. The one-vertex sequence (v) is an uphill path if and only if v is a valley, and a valley has no lower neighbours. The inequality $N(v) \geq 1$ follows by induction on $f(v)$: a non-valley has at least one lower neighbour. $\square$

A.2 Vertices with a single uphill path

Put $S = S_f = \{v : N(v) = 1\}$ and $R = R_f = \{v : N(v) \geq 2\}$. Write $G[X]$ for the subgraph induced by X , and $e(X)$ for its number of edges.

Lemma A.2. (a) *If w is a lower neighbour of v , then $N(v) \geq N(w)$. A vertex of S that is not a valley has exactly one lower neighbour, and it lies in S .*

(b) *Every edge between $s \in S$ and $u \in R$ satisfies $f(s) < f(u)$. Every higher neighbour of a vertex of R lies in R .*

(c) *$G[S]$ is a forest. Every valley of G lies in S , and every component of $G[S]$ contains exactly one valley.*

Proof. (a) By Lemma A.1, $N(v)$ is a sum of terms ≥ 1 that includes $N(w)$. If $v \in S$ is not a valley, then this sum equals 1, so it has a single term, equal to 1.

(b) If $f(u) < f(s)$, then u is a lower neighbour of s and (a) gives $N(s) \geq N(u) \geq 2$, a contradiction. Likewise, a higher neighbour v of $u \in R$ has $N(v) \geq N(u) \geq 2$.

(c) Valleys have $N = 1$. Every edge of $G[S]$ has a higher end v , and by (a) the lower end is the unique lower neighbour of v . So $e(S)$ equals the number of non-valleys in S . Let K be a component of $G[S]$, with n_K vertices. Its vertex of smallest label has no lower neighbour in K . By (a) it has no lower neighbour in G at all, so it is a valley. Counting as above inside K , $e(K) = n_K - (\text{number of valleys in } K) \leq n_K - 1$. A connected graph has $e(K) \geq n_K - 1$. Hence K is a tree and contains exactly one valley. $\square$

A.3 An exact formula

Proposition A.3 (Exact formula). *For every graph G and every labelling f ,*

$$P(f) = |V(G)| + \sum_{u \in R} (\deg u - 1) + \sum_{u \in R} (N(u) - 2) \text{out}(u).$$

All terms of the last sum are ≥ 0 . In particular, if G is d -regular, then $P(f) \geq |V(G)| + (d-1)|R_f|$.

Proof. Sum the recursion of Lemma A.1 over all v . Each pair $(w, \text{higher neighbour } v \text{ of } w)$ contributes $N(w)$ once, so

$$P(f) = \#\{\text{valleys}\} + \sum_w N(w) \text{out}(w).$$

Let c be the number of components of $G[S]$. By Lemma A.2(c), there are c valleys and $e(S) = |S| - c$. For $w \in S$ we have $N(w) = 1$. Moreover, $\sum_{w \in S} \text{out}(w) = e(S) + e(S, R)$, because each edge inside

S is counted at its lower end, each S – R edge is counted at its end in S (Lemma A.2(b)), and edges inside R are not counted. For $w \in R$, all higher neighbours lie in R (Lemma A.2(b)), so $\sum_{w \in R} \text{out}(w) = e(R)$. Using $e(S, R) = \sum_{u \in R} \deg u - 2e(R)$, we get

$$P(f) = c + (|S| - c) + e(S, R) + \sum_{u \in R} N(u) \text{out}(u) = |S| + \sum_{u \in R} \deg u + \sum_{u \in R} (N(u) - 2) \text{out}(u),$$

and $|S| = |V(G)| - |R|$. Finally $N(u) \geq 2$ on R . $\square$

A.4 Decycling sets

A set $R \subseteq V(G)$ is a *decycling set* (or feedback vertex set) if $G - R = G[V \setminus R]$ is a forest. The *decycling number* $\nabla(G)$ is the smallest size of a decycling set. By Lemma A.2(c), R_f is a decycling set for every labelling f .

Theorem A.4 (Lower bound). *If G is d -regular, then $U(G) \geq |V(G)| + (d-1)\nabla(G)$. In particular, $U(Q_d) \geq 2^d + (d-1)\nabla(Q_d)$ for every $d \geq 1$.*

Proof. Proposition A.3 and $|R_f| \geq \nabla(G)$. $\square$

Proposition A.5 (Upper bound). *Let G be d -regular and let R be a decycling set that is independent (no edge inside R). Then some labelling has exactly $|V(G)| + (d-1)|R|$ uphill paths. Consequently, if Q_d has an independent decycling set of size $\nabla(Q_d)$, then $U(Q_d) = 2^d + (d-1)\nabla(Q_d)$.*

Proof. Give the smallest labels to $V \setminus R$, tree by tree. Inside each tree of the forest $G - R$, use breadth-first order from a root. Then give the remaining labels to R , in any order. The neighbours of a vertex $v \notin R$ are its tree neighbours and some vertices of R , and the latter come later. In breadth-first order, a non-root vertex comes after its parent and before its children. So a root is a valley, and a non-root vertex has exactly one lower neighbour, its parent. By induction along the order, $N(v) = 1$ for all $v \notin R$. A vertex $u \in R$ has all its d neighbours outside R , all labelled earlier, so $N(u) = d$. The total is $(|V| - |R|) + d|R|$. The last sentence then follows from Theorem A.4. $\square$

A.5 Code labellings

For $x \in \{0, 1\}^d$, let $|x|$ be its weight (number of ones). Let E_d and O_d be the sets of vertices of even and of odd weight. Every edge of Q_d joins E_d to O_d , so both sets are independent and have size 2^{d-1} . The distance $|x \oplus y|$ of two vertices of E_d is even.

Proposition A.6 (Code labelling). *Let $d \geq 2$, and let $C \subseteq E_d$ have pairwise distances ≥ 4 . Consider the labelling L_C that lists first the vertices of C , then all of O_d , then $E_d \setminus C$ (each block in any order). It has exactly*

$$P(L_C) = 2^d + (d-1)(2^{d-1} - |C|)$$

uphill paths. Moreover, $E_d \setminus C$ is an independent decycling set of Q_d . Hence $\nabla(Q_d) \leq 2^{d-1} - |C|$ and $U(Q_d) \leq 2^d + (d-1)(2^{d-1} - |C|)$.

Proof. We use Lemma A.1. A codeword has only odd neighbours, all labelled later, so it is a valley: $N = 1$. The earlier neighbours of an odd vertex x are exactly its neighbours in C . Two of them would be at distance 2 from each other, so there is at most one. Hence $N(x) = 1$: either x is a valley, or it has a single lower neighbour, with $N = 1$. A vertex $y \in E_d \setminus C$ has d odd neighbours, all labelled earlier, so $N(y) = d$. Summing, $P = |C| + 2^{d-1} + d(2^{d-1} - |C|)$. In this labelling $R_{L_C} = E_d \setminus C$ (as $d \geq 2$), which is decycling by Lemma A.2(c) and independent since it lies in E_d . $\square$

A.6 A counting bound for the decycling number

Lemma A.7. *Let G be d -regular with n vertices and m edges, and let $R \neq V(G)$ be a decycling set. Let c be the number of components of $G - R$. Then*

$$(d-1)|R| = m - n + c + e(R) \geq m - n + 1.$$

For Q_d this reads $(d-1)|R| \geq (d-2)2^{d-1} + 1$.

Proof. $G - R$ is a forest with $n - |R|$ vertices and c components, so it has $n - |R| - c$ edges. The edges meeting R number $d|R| - e(R)$. Adding up gives $m = n - |R| - c + d|R| - e(R)$. $\square$

A.7 Q_3 and Q_4

Theorem A.8. $U(Q_3) = 14$ and $U(Q_4) = 34$. Also $\nabla(Q_3) = 3$ and $\nabla(Q_4) = 6$.

Proof. $d = 3$: Lemma A.7 gives $2\nabla(Q_3) \geq 5$, so $\nabla(Q_3) \geq 3$, and Theorem A.4 gives $U(Q_3) \geq 8 + 2 \cdot 3 = 14$. The code $C = \{000\}$ in Proposition A.6 gives a labelling with $8 + 2 \cdot (4 - 1) = 14$ uphill paths and a decycling set of size 3.

$d = 4$: Lemma A.7 gives $3\nabla(Q_4) \geq 17$, so $\nabla(Q_4) \geq 6$ and $U(Q_4) \geq 16 + 3 \cdot 6 = 34$. The code $C = \{0000, 1111\}$ (distance 4) gives $16 + 3 \cdot (8 - 2) = 34$ and a decycling set of size 6. $\square$

Remark A.9. An exhaustive check over all $8! = 40\,320$ labellings of Q_3 (`uphill_eval.py --brute 3`) also returns 14. The proof above does not use it. For $d = 2$ the same argument gives $U(Q_2) = U(C_4) = 5$. In words, Proposition A.3 says that an optimal labelling must make few vertices “expensive” ($N \geq 2$), and those vertices must still break every cycle. For an arbitrary graph, the edge count in the proof of Lemma A.7 reads $|E| = |S| - c + \sum_{u \in R} \deg u - e(R)$. Combined with Proposition A.3, it gives $P(f) \geq |E| + c + e(R) \geq |E| + 1$. This is the lower bound $2n^2 - 2n + 1$ of the grid problem [3].

A.8 The labellings

Vertices in increasing label order (the number before each row is the label of its first entry). The labellings are L_C of Proposition A.6: first the codewords, then the odd vertices, then the remaining even vertices, each block in increasing binary order.

Q_3 , $C = \{000\}$, 14 uphill paths:

1: 000 001 010 100 111 011 101 110

Q_4 , $C = \{0000, 1111\}$, 34 uphill paths:

1: 0000 1111 0001 0010 0100 0111 1000 1011

9: 1101 1110 0011 0101 0110 1001 1010 1100

Both counts were confirmed with the exact evaluator `code/uphill_eval.py`, which applies Lemma A.1 directly.

References

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